

CSE 106: Online on Graph

Section: B

Duration: 20 Minutes

Background

You are analyzing a social media network consisting of its users. For a specific person S , some people in the network are directly connected to S (friends), while others are not (strangers). Your task is to determine how many mutual friends S has with each of these strangers.

A mutual friend between person A and another person B is a person M such that, M is directly friends with A and B . In other words, a mutual friend is a common friend shared by A and B . In the figure, Elisa is directly connected to Anne and Carl, so they are her friends. Bob and Diana are not directly connected to Elisa, so they are considered strangers to her. Among the strangers, Diana is connected to both Anne and Carl, and since Anne and Carl are also Elisa's friends, they are mutual friends of Elisa and Diana. Similarly, for the Elisa, Bob stranger pair, the mutual friends are Anne, Carl.

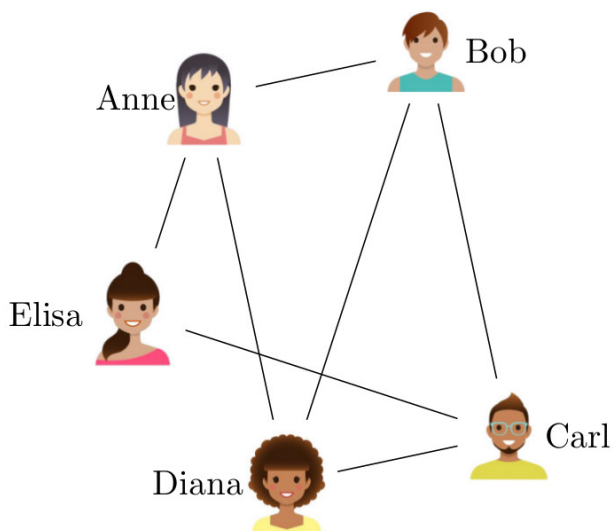


Figure 1: A sample friendship network

Input Format:

- The first line of input contains a single integer M , representing the total number of friendship pairs in the network.

- Each of the next M lines contains a pair written in the form [A, B], indicating that person A and person B are friends.
- The last line contains the person of interest, i.e., the person for whom you have to find the mutual friends with all the users who are strangers to him.

Output Format:

- For each stranger X, print one line in the format: $S - - X : K$ where:
 S is the person of interest
 X is a user stranger to S and
 K is the number of mutual friends between S and X

Sample I/O

Example - 1:

Input:

```
5
[A, B]
[B, C]
[E, D]
[A, E]
[C, F]
A
```

Output:

```
A -- C: 1
A -- D: 1
A -- F: 0
```

Example - 2:

Input:

```
8
[P, Q]
[P, R]
[Q, R]
[Q, S]
[R, T]
[S, T]
[T, P]
[U, V]
P
```

Output:

P -- S: 2

P -- U: 0

P -- V: 0